# CEE 260/MIE 273: Probability and Statistics in Civil Engineering
# Lecture 5A: Inference for Single Proportion

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# Outline

1. Test for normality
2. CI for proportion
3. Hypothesis testing
4. Using p-values
5. Outlook

# Hypothesis testing using *p*-value approach

[redacted]

Step 5. Calculate test statistic value:

$$z = \frac{.723 - .5}{\sqrt{.5(.5)/400}} = 8.92$$

Step 6. Determine *p*-value (two-tailed test):

$$p\text{-value} = 2(1 - \Phi(8.92)) = 0.0$$

Step 7. Compare *p*-value to $\alpha = 0.01$:

$$0.0 \leq 0.01$$

Step 8. Conclude:
Using a significance level of 0.01, we reject $H_0$ since $0.0 < 0.01$.

# Example 5: Equipment quality (p-value approach)

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A car manufacturer is considering switching to a new, higher quality piece of equipment that con- structs vehicle door hinges. They figure that they will save money in the long run if this new machine produces hinges that have flaws no more than 0.2% of the time. A random sample of 800 hinges produced by the new machine shows that 4 are flawed. At the 0.05 significance level, is there enough evidence to conclude that the new machine produces hinges with a flaw rate greater than 0.2%?

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**Test Statistic:**

$$\hat{p} = \frac{4}{800} = 0.005$$

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**Test Statistic:**

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\begin{aligned}
\hat{p} &= \frac{4}{800} = 0.005 \\
SE &= \sqrt{\frac{0.002(1-0.002)}{800}} \approx 0.0016
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**p-value:** For $z = 1.88$, the $p$-value is:

$$p\text{-value} = 1 - \Phi(1.88) \quad [\texttt{1 - norm.cdf(1.88)}]$$

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&= \approx 0.0304
\end{aligned}$$

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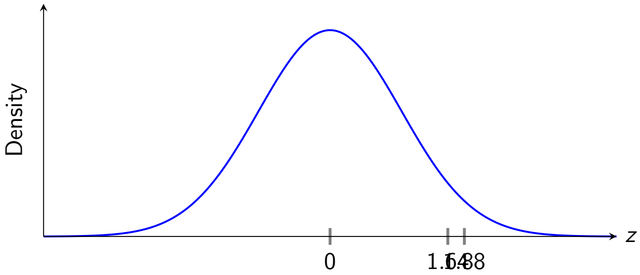

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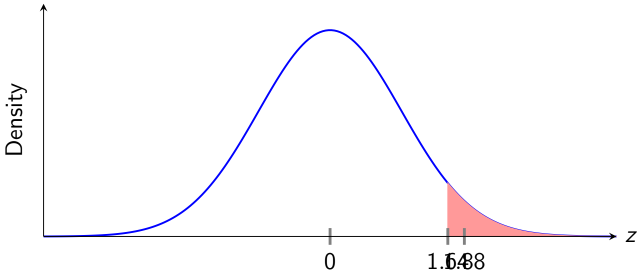

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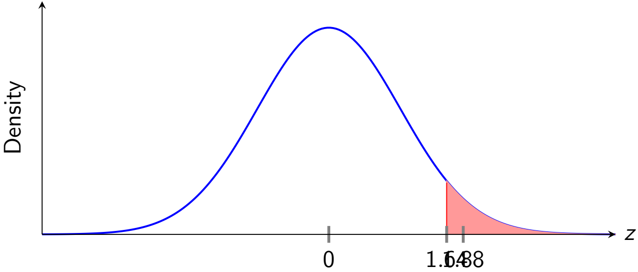

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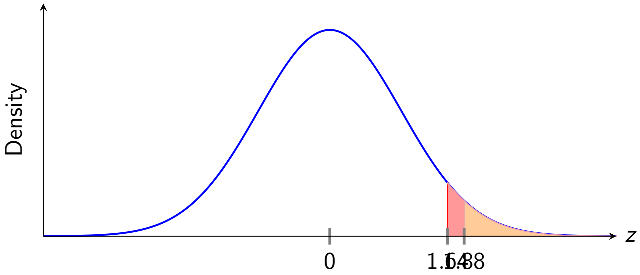

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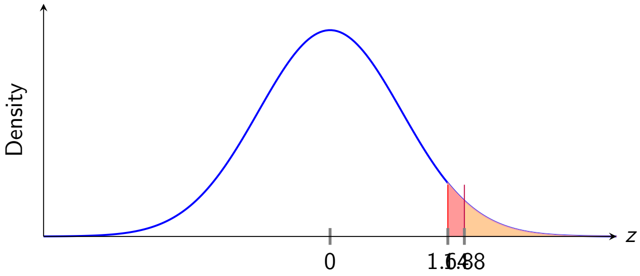

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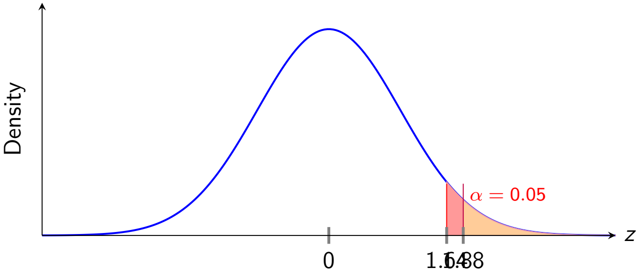

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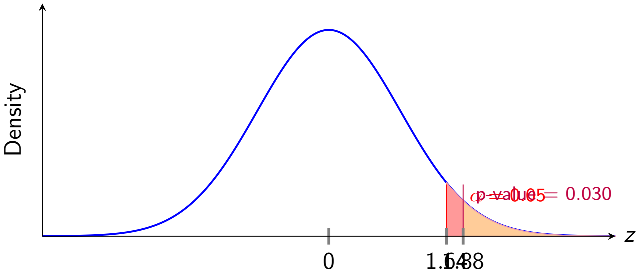

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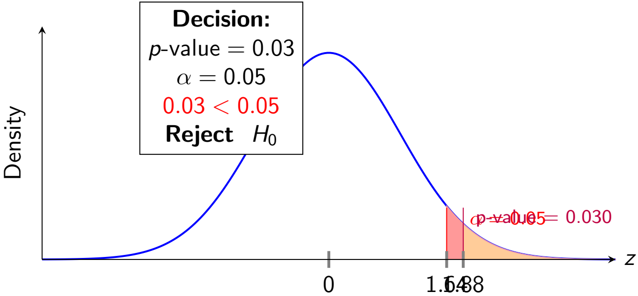

**Decision:**
$p$-value $= 0.03$
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**Reject** $H_0$

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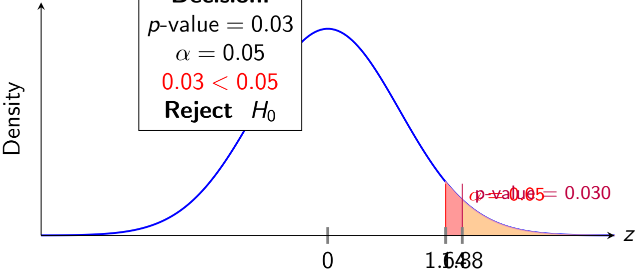

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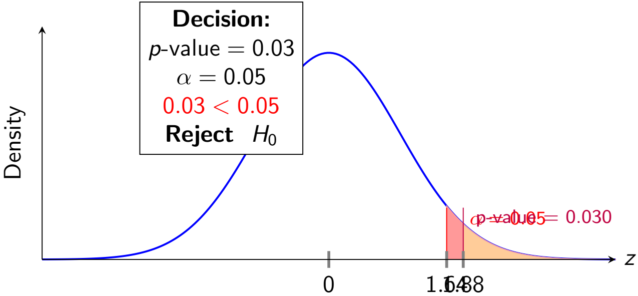

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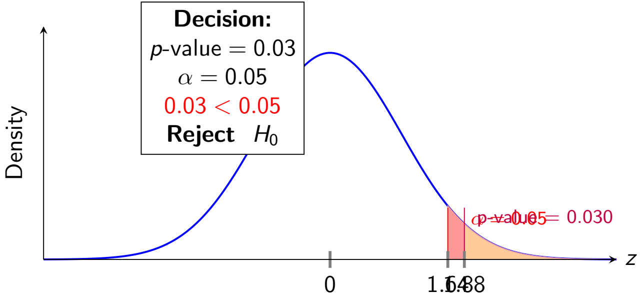

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**Conclusion:** There is sufficient evidence to conclude that the new machine produces hinges with a flaw rate greater than 0.2%.

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